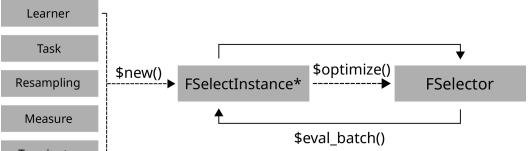


Feature Selection with mlr3fselect::CHEAT SHEET

Class Overview

The package provides a set of R6 classes which allow to (a) define general feature selection instances; (b) run black-box optimizers; (c) combine learners with feature selection (for nested resampling).



[NB: In many table prints we suppress cols for readability.]

Terminators - When to stop

Construction: `trm(.key, ...)`

- `evals(n_evals)`
After iterations.
- `run_time(secs)`
After training time.
- `clock_time(stop_time)`
At given timepoint.
- `perf_reached(level)`
After performance was reached.
- `stagnation(iters, threshold)`
After performance stagnated.
- `stagnation_batch(n, threshold)`
After performance stagnated for batches.
- `combo(list_of_terms, any=TRUE)`
Combine terminators with AND or OR.

```
as.data.table(mlr_terminators) # list all
```

Lists all available terminators.

FSelectInstance* - Search Scenario

Evaluator and container for resampled performances of feature subsets. The (internal) function `eval_batch(xdt)` calls `benchmark()` to evaluate a table of feature subsets. Stores archive of all evaluated feature subsets and the final result.

```
instance = FSelectInstanceBatchSingleCrit$new(task, learner, resampling, measure, terminator)
```

`store_benchmark_result = TRUE` to store resampled evals and `store_models = TRUE` for fitted models.

Example

```
instance = FSelectInstanceBatchSingleCrit$new(task, learner, resampling, measure, terminator)
fselector = fs("random_search", batch_size = 10)
fselector$optimize(instance)
instance$result
# >   Petal.Length Petal.Width Sepal.Length Sepal.Width classif.ce
# > 1:      FALSE      TRUE      TRUE      TRUE      0.06
```

Use `FSelectInstanceBatchMultiCrit` for multi-criteria feature selection.

FSelector - Search Strategy

Generates feature subsets and passes to instance for evaluation until termination. Creation: `fs(.key, ...)`

- `random_search(batch_size)`
Random search.
- `exhaustive_search(max_features)`
Exhaustive Search.
- `sequential(strategy)`
Sequential Selection.
- `rfe(feature_fraction, recursive)`
Recursive Feature Elimination.
- `design_points(batch_size, design)`
User supplied feature subsets.

```
as.data.table(mlr_fselectors) # list all
```

Lists all available feature selection algorithms.

Logging and Parallelization

```
lgr::get_logger("bbotk")$set_threshold("<level=")
```

Change log-level only for `mlr3fselect`.

```
future::plan(strategy)
```

Sets the parallelization backend. Speeds up feature selection by running iterations in parallel.

Executing the Feature Selection

```
fselector$optimize(instance)
as.data.table(instance$archive)
## >   Petal.Length Petal.Width Sepal.Length Sepal.Width classif.ce
## > 1:      TRUE      TRUE      TRUE      TRUE      0.09333333
## > 2:      TRUE      TRUE      TRUE      FALSE      0.09333333
instance$result # datatable row with optimal feature subset and estimated perf
```

Get evaluated feature subsets and performances; and result.

```
task_selected = task$clone(deep = TRUE)
task_selected$select(instance$result_feature_
```

Set optimized feature subset in a cloned Task.

Example

```
instance = fselect(fs("random_search"), task = tsk("iris"), learner = learner, resampling = rsm("holdout"), measures = msr("classif.ce"), term_evals = 20)
```

Use `fselect()`-shortcut.

AutoFSelector - Select before Train

Wraps learner and performs integrated feature selection.

```
afs = AutoFSelector$new(fselector = fselector, learner = learner, resampling = resampling, measure = measure, terminator = terminator)
```

Inherits from class `Learner`. Training starts feature selection on the training set. After completion the learner is trained with the "optimal" feature subset on the given task.

```
afs$train(task)
afs$predict(task, row_ids)
```

```
afs$learner
```

Returns learner trained on full data set with optimized feature subset.

```
afs$fselect_result
# >   Petal.Width Sepal.Length Sepal.Width classif.ce
# > 1:      TRUE      TRUE      TRUE      0.02
```

Access feature selection result.

```
afs = auto_fselector(fs("random_search"), learner, resampling, measure, term_evals = 20)
```

Use shortcut to create `AutoFSelector`.

Nested Resampling

Just resample `AutoFSelector`; now has inner and outer loop.

Example

```
inner = rsm("holdout")
afs = auto_fselector(fs("random_search"), learner, inner, measure, term_evals = 20)
outer = rsm("cv", folds = 2)
rr = resample(task, afs, outer, store_models = TRUE)

as.data.table(rr)
## >   iteration learner resampling iteration prediction
## > 1:   <AutoFSelector[38]>   <ResamplingCV[19]>      1   <PredictionClassif[19]>
## > 2:   <AutoFSelector[38]>   <ResamplingCV[19]>      2   <PredictionClassif[19]>
```

```
extract_inner_fselect_results(rr)
# >   iteration Petal.Width Sepal.Length Sepal.Width classif.ce
# > 1:      1      TRUE      TRUE      TRUE      0.04
# > 2:      2      TRUE      TRUE      FALSE      0.00
```

Check inner results for stable features.

```
rr$score()
# >   learner iteration prediction classif.ce
# > 1: <AutoFSelector[40]>      1 <PredictionClassif[19]> 0.02666667
# > 2: <AutoFSelector[40]>      2 <PredictionClassif[19]> 0.08000000
```

Predictive performances estimated on the outer resampling.

```
extract_inner_fselect_archives(rr)
# >   iteration Petal.Width Sepal.Length Sepal.Width classif.ce
# > 1:      1      FALSE      TRUE      FALSE      0.36
# > 21:      2      FALSE      TRUE      FALSE      0.44
```

All evaluated feature subsets.

```
rr$aggregate()
# > classif.ce
# > 0.05333333
```

Aggregates performances of outer resampling iterations.

```
rr = fselect_nested(fs("random_search"), task, learner, inner, outer, measure, term_evals = 20)
```

Use shortcut to execute nested resampling.